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FED-STD-311A/GEN
04 November 2024
SUPERSEDING
FED-STD-311
15 January 1969

FEDERAL TEST METHOD STANDARD

LEATHER, METHODS OF SAMPLING AND TESTING

The General Services Administration has authorized the use of this federal standard by all federal agencies.

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Preamble

The Acquisition Reform Act of 1994 established the policy for the future procurements of Military material. This policy emphasizes the use of commercial items, practices, processes, and standards, to the maximum extent, in lieu of federal and military specifications, and standards.

This Federal Standard is updated to include commercial, non-government test methods which have been identified as direct conversions to those cited in Federal Test Method Standard No. 311.

FED-STD-311A was prepared by DLA Troop Support, Clothing & Textiles and is an update of FED-STD-311 dated January 15, 1969, and modified by Change Notices 1, 2, and 3.

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FEDERAL TEST METHOD STANDARD

LEATHER, METHODS OF SAMPLING AND TESTING

1. SCOPE

1.1. Scope. This standard describes the general methods for testing leather and leather products for conformance with the requirements of Federal and Military Specifications. In the event that conflict should occur between the requirements of this standard and those contained in a material specification or other type of procurement document on date of invitation for bid, the procurement document or specification shall govern.

1.2. Applicability. The methods of sampling and testing cited in this standard can be used by all Services for testing and sampling of leather materials used in footwear, handwear and other leather products.

2. INDEX OF TEST METHODS

2.1. Alphabetical Index of Test Methods.

Title	Method No.	Status: Canceled or Superseded by
Acetone Insoluble Material	6371	Canceled
Aldehyde, Qualitative	6111	Canceled
Alkali-Solubility of Leather-Cellulose Mixtures	6431	Canceled
Apparent Density of Leather	1231	ASTM D2346
Area Stability of Leather	7031	Canceled
Area Stability of Leather to Laundering	3041.1	ASTM D6013 2/
Area Stability of Leather to Perspiration	3211.2	ASTM D2322 3/
Area, Unit, Template Method	1121	Canceled
Area, Unit, Weight Method	1111	Canceled
Ash, Total	6421	ASTM D2617
Blocking	3121.1	ASTM D6116
Breaking Force and Elongation, Grab Method	2031	ASTM D2208
Breaking Force or Tensile Strength, and Elongation: Dumbbell Strip Method	2021	ASTM D2209
Bursting Strength, Plunger Method	2051	ASTM D2207
Chemical Tests, General	6001	ASTM D2813
Chlorides, Soluble	6351	Canceled
Chloroform Extraction	6311	Canceled
Chloroform-Soluble and Water-Soluble Materials, and Insoluble Inorganic Materials	6341	ASTM D3495
Chromic Oxide, Colorimetric Method	6521	Canceled
Chromic Oxide, Fusion Method	6511	Canceled

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Title	Method No.	Status: Canceled or Superseded by
Chromic Oxide, Perchloric Acid Method	6515	ASTM D2807
Chromic Oxide, Wet Oxidation	6518	Canceled
Colorfastness (Resistance to Rubbing), AATCC Crockmeter	3011	ASTM D5053 1/
Colorfastness (Resistance to Rubbing), Precision Crockmeter	3031.1	ASTM D5053 1/
Combined Tannin and Degree of Tannage	6631	ASTM D6020
Compressibility	4411	ASTM D2213
Corrosive Effects of Leather	3203	ASTM D1611
Cracking, Hand	4021	ASTM D6075
Cracking, Low Temperature	7211	ASTM D6075
Cracking, Vise	4011	ASTM D6075
Dynamic Absorption of Leather Surfaces	8151	ASTM D6014
Dynamic Water Resistance of Leather by Maeser Water Penetration Tester	8021.1	ASTM D2099
Dynamic Water Resistance of Shoe Upper Leather by the Dow Corning Leather Tester	8131.1	ASTM D2098
Finish Stability	3202	ASTM D7912
Gasoline-Water Separation	6811	Canceled
Iron and Aluminum Oxides	6531	Canceled
Lead Salts, Presence of	6551	ISO 17072-2
Magnesium Sulfate (Epsom Salt)	6541	ASTM D6017
Mildew Resistance; Direct Inoculation, Pure Culture, Non-Sterile Specimen Method	5021	Canceled
Mildew Resistance, Tropical Chamber Method	5041	Canceled
Moisture Content, Dry Air Method	6211	Canceled
Moisture Content, Oven Method	6221	ASTM D3790
Nitrogen, Collagenous and Hide Substance	6441	ASTM D2868
Nitrogen, Water Extractable	6452	ASTM D6016
Paranitrophenol, Colorimetric Method	6711	Canceled
Permeability, Water Vapor	8011.2	ASTM D5052
pH Value of Leather	6621.1	ASTM D2810
Piping	4111	ASTM D2941
Preparation of Composite Sample	6002.1	ASTM D2813
Resistance to Grain Crack, Mullen Test	2211	ASTM D2210
Resistance to Perspiration (White Chrome Leather)	3221	Canceled
Shear Strength, Stuck-on Rib	2061	Canceled
Shrinkage Temperature	7011.1	ASTM D6076
Staining	3021.2	ASTM D6012 1/
Stiffness	4211.2	ASTM D2821
Sulfates, Soluble	6361	Canceled

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Title	Method No.	Status: Canceled or Superseded by
Tackiness	3111	ASTM D6183
Tearing Strength, Buckle	2131	ASTM D4831
Tearing Strength, Split	2121	Canceled
Tearing Strength, Slot Tear	2161	ASTM D2212
Tearing Strength, Stitch, Double Hole	2151	ASTM D4705
Tearing Strength, Stitch, Single Hole	2141	ASTM D4786
Tearing Strength, Stitch, Stuck-on	2171	Canceled
Tearing Strength, Tongue	2111	ASTM D4704
Thickness, Specimen	1021.1	ASTM D1813
Thickness, Unit	1011.1	ASTM D1814
Torsion Wire Stiffness Measurement	4212	ASTM D2821
Water Absorption, Static Method	8111.1	ASTM D6015
Water Absorption, Tumble Method	8112.1	Canceled
Water Absorption, Insoles	8141	ASTM D6014
Water Marks on Grain Leather	8231	Canceled
Water Resistance, with Tap Tester	8121	Canceled
Width and Length, Unit	1211	ASTM D1516

1/ Rated using AATCC EP8 – Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale.
2/ 1991 AATCC Reference Detergent without optical brightener may be used in place of the soap chips.
3/ Except vacuum soaking shall be used to effect perspiration immersion. The suction flask shall contain 15 ± 1 mL of artificial perspiration solution per specimen and the circulating air oven shall be preheated to 68 ± 2 °C.

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2.2. Numerical Index of Test Methods.

GROUP 1000 – DIMENSIONAL MEASUREMENTS

Title	Method No.	Status: Canceled or Superseded by
Thickness, Unit	1011.1	ASTM D1814
Thickness, Specimen	1021.1	ASTM D1813
Area, Unit, Weight Method	1111	Canceled
Area, Unit, Template Method	1121	Canceled
Width and Length, Unit	1211	ASTM D1516
Apparent Density of Leather	1231	ASTM D2346

GROUP 2000 – STRENGTH TESTS

Title	Method No.	Status: Canceled or Superseded by
Breaking Force or Tensile Strength, and Elongation: Dumbbell Strip Method	2021	ASTM D2209
Breaking Force and Elongation, Grab Method	2031	ASTM D2208
Bursting Strength, Plunger Method	2051	ASTM D2207
Shear Strength, Stuck-on Rib	2061	Canceled
Tearing Strength, Tongue	2111	ASTM D4704
Tearing Strength, Split	2121	Canceled
Tearing Strength, Buckle	2131	ASTM D4831
Tearing Strength, Stitch, Single Hole	2141	ASTM D4786
Tearing Strength, Stitch, Double Hole	2151	ASTM D4705
Tearing Strength, Slot Tear	2161	ASTM D2212
Tearing Strength, Stitch, Stuck-on Rib	2171	Canceled
Resistance to Grain Crack, Mullen Test	2211	ASTM D2210

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GROUP 3000 – PERFORMANCE TESTS

Title	Method No.	Status: Canceled or Superseded by
Staining	3021.2	ASTM D6012 1/
Colorfastness (Resistance to Rubbing), AATCC Crockmeter	3011	ASTM D5053 1/
Colorfastness (Resistance to Rubbing), Precision Crockmeter	3031.1	ASTM D5053 1/
Area Stability of Leather to Laundering	3041.1	ASTM D6013 2/
Tackiness	3111	ASTM D6183
Blocking	3121.1	ASTM D6116
Finish Stability	3202	ASTM D7912
Corrosive Effects of Leather	3203	ASTM D1611
Area Stability of Leather to Perspiration	3211.2	ASTM D2322 3/
Resistance to Perspiration (White Chrome Leather)	3221	Canceled

1/ Rated using AATCC EP8 – Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale.
2/ 1991 AATCC Reference Detergent without optical brightener may be used in place of the soap chips.
3/ Except vacuum soaking shall be used to effect perspiration immersion. The suction flask shall contain 15 ± 1 mL of artificial perspiration solution per specimen and the circulating air oven shall be preheated to 68 ± 2 °C.

GROUP 4000 – MECHANICAL TESTS

Title	Method No.	Status: Canceled or Superseded by
Cracking, Vise	4011	ASTM D6075
Cracking, Hand	4021	ASTM D6075
Piping	4111	ASTM D2941
Stiffness	4211.2	ASTM D2821
Torsion Wire Stiffness Measurement	4212	ASTM D2821
Compressibility	4411	ASTM D2213

GROUP 5000 – BIOLOGICAL TESTS

Title	Method No.	Status: Canceled or Superseded by
Mildew Resistance; Direct Inoculation, Pure Culture, Non-Sterile Specimen Method	5021	Canceled
Mildew Resistance, Tropical Chamber Method	5041	Canceled

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GROUP 6000 – CHEMICAL TESTS

Title	Method No.	Status: Canceled or Superseded by
Chemical Tests, General	6001	ASTM D2813
Preparation of Composite Sample	6002.1	ASTM D2813
Aldehyde, Qualitative	6111	Canceled
Moisture Content, Dry Air Method	6211	Canceled
Moisture Content, Oven Method	6221	ASTM D3790
Chloroform Extraction	6311	Canceled
Chloroform-Soluble and Water-Soluble Materials, and Insoluble Inorganic Materials	6341	ASTM D3495
Chlorides, Soluble	6351	Canceled
Sulfates, Soluble	6361	Canceled
Acetone Insoluble Material	6371	Canceled
Ash, Total	6421	ASTM D2617
Alkali-Solubility of Leather-Cellulose Mixtures	6431	Canceled
Nitrogen, Collagenous and Hide Substance	6441	ASTM D2868
Nitrogen, Water Extractable	6452	ASTM D6016
Chromic Oxide, Fusion Method	6511	Canceled
Chromic Oxide, Perchloric Acid Method	6515	ASTM D2807
Chromic Oxide, Wet Oxidation	6518	Canceled
Chromic Oxide, Colorimetric Method	6521	Canceled
Iron and Aluminum Oxides	6531	Canceled
Magnesium Sulfate (Epsom Salt)	6541	ASTM D6017
Lead Salts, Presence of	6551	ISO 17072-2
pH Value of Leather	6621.1	ASTM D2810
Combined Tannin and Degree of Tannage	6631	ASTM D6020
Paranitrophenol, Colorimetric Method	6711	Canceled
Gasoline-Water Separation	6811	Canceled

GROUP 7000 – THERMAL TESTS

Title	Method No.	Status: Canceled or Superseded by
Shrinkage Temperature	7011.1	ASTM D6076
Area Stability of Leather	7031	Canceled
Cracking, Low Temperature	7211	ASTM D6075

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GROUP 8000 – WATER RESISTANCE TESTS

Title	Method No.	Status: Canceled or Superseded by
Permeability, Water Vapor	8011.2	ASTM D5052
Dynamic Water Resistance of Leather by Maeser Water Penetration Tester	8021.1	ASTM D2099
Water Absorption, Static Method	8111.1	ASTM D6015
Water Absorption, Tumble Method	8112.1	Canceled
Water Resistance, with Tap Tester	8121	Canceled
Dynamic Water Resistance of Shoe Upper Leather by the Dow Corning Leather Tester	8131.1	ASTM D2098
Water Absorption, Insoles	8141	ASTM D6014
Dynamic Absorption of Leather Surfaces	8151	ASTM D6014
Water Marks on Grain Leather	8231	Canceled

3. DEFINITION OF TERMS

3.1. Scope. This section covers the definitions of terms used in the leather industry, including some of those encountered by personnel concerned with specifications and procurement of hides and leather products by the Federal Government. Terms adequately described by unabridged dictionaries are not included.

3.2. Definitions. Definitions for the terms used in the leather industry shall be in accordance with ASTM D1517 Standard Terminology Relating to Leather.

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4. SAMPLING FOR INSPECTION

4.1. Scope. This section establishes sampling plans for inspection by attributes of leather and fabricated leather articles. The term "inspection" is used throughout the specification to signify the process of measuring, examining, testing, or otherwise comparing the unit of product. The term "sampling" is used to signify lot-by-lot acceptance sampling. The essential feature is that a product is grouped into lots, each of which is accepted or rejected in its entirety on the basis of the performance of a sample or samples taken from it.

4.2. Quality assurance terms and definitions. Definitions for quality assurance terms shall be in accordance with ASQ/ANSI Z1.4, "Sampling Procedures and Tables for Inspection by Attributes".

4.3. General sampling procedure.

4.3.1. Formation of lots or batches. Prior to sampling, the product shall be assembled into identifiable lots, batches, or any such other manner as may be specified in the applicable procurement document. Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions. Unless otherwise specified in the material specification, the number of units of product comprising a lot shall not exceed the equivalent of 25,000 square feet of leather.

4.3.1.1. The leather in any one lot should be produced:

(a) From units of product of similar size and type.

(b) From functionally equivalent tanning and finishing materials.

(c) From a single production method.

(d) From sequential production batches.

4.3.1.2. Presentation of lots or batches. The formation of the lots or batches, the lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier will provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentations, and personnel for all handling of product required for drawing of samples.

4.4. Sampling for inspection. The sample required for inspection of the lot shall be taken in accordance with a random process, i.e., a process which gives each item in the lot the same chance of being included in the sample. To this end, the sample procedure should conform as far as practicable to the following rules:

Rule 1: Units shall be taken from locations scattered throughout the lot to ensure that all the units for a sample will not be taken from the same portion of the lot, such as a single carton, layer, tier, etc.

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Rule 2: Units shall be taken without regard to quality.

4.5. Sampling for examination of visual, dimensional, and tactile characteristics.

4.5.1. Sample size. Unless otherwise specified in the material specification, the number of units to be taken from a lot for the purpose of visual, dimensional, or tactile characteristic examination shall be in accordance with Table I.

TABLE I. Sampling for examination of visual, dimensional, and tactile characteristics of nonfabricated and fabricated leather articles.

Lot size	Sample size	Rejection Number	
		Normal	Tightened
8 or less	2	1	1
9 to 15	3	1	1
16 to 25	5	1	1
26 to 50	8	1	1
51 to 90	13	2	2
91 to 150	20	2	2
151 to 280	32	3	2
281 to 500	50	4	3
501 to 1200	80	6	4
1201 to 3200	125	8	6
3201 to 10000	200	11	9
10001 to 35000	315	15	13
35001 and over	500	22	19

4.5.2. Normal inspection. Normal inspection shall be used at the start of inspection, unless otherwise directed by the procuring activity. When normal inspection is in effect, tightened inspection shall be instituted when two out of five consecutive lots or batches have been rejected on original inspection.

4.5.3. Tightened inspection. When tightened inspection is in effect, normal inspection shall be instituted when five consecutive lots or batches have been considered acceptable on original inspection.

4.5.4. Rejection of lot on the basis of examination of the sample. A lot should not be tested for physical and chemical characteristics if the number of defective units in the sample equals or exceeds the “Rejection Number.” If the number of defective units in the sample is less than the “Rejection Number,” the lot should be sampled for physical and chemical characteristics to determine its acceptability.

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4.6. Sampling for physical and chemical tests. The sample required for testing the lot for physical and chemical characteristics shall be selected from the sample for visual, dimensional, or tactile characteristics. Each item of the sample, regardless of its visual, dimensional, or tactile characteristics, shall be given an equal chance of being included in the sample for testing.

4.6.1. Sampling for testing of nonfabricated leather.

4.6.1.1. Sample size. Unless otherwise specified in the material specification, 15 units of product shall be selected at random from each lot for the purpose of testing. For lots consisting of less than 15 units of product, each unit shall be sampled for test.

4.6.1.2. Location of test area. A 12 by 12-inch area of the unit of product from which pieces shall be taken for use in preparing specimens for physical and chemical tests in shown in Figure 1. The piece taken from the test area shall be of sufficient size and shape to furnish all the specimens required by the material specification for tests. The specimens shall be taken from the 12- by 12-inch test area as close to point "a" as is practicable. The piece shall be marked to indicate the side that is parallel to the backbone. The size and shape of the specimen for test cut from the piece shall be as specified in the applicable test method or in the material specification. Unless otherwise specified in the applicable test method or in the detail specification, the specimen for test shall be cut with the long dimension perpendicular to the backbone.

4.6.1.2.1. Skins. The test area shall correspond to that for cattlehides as shown in Figure 1, except that it shall begin at a point 1 inch from the backbone line and 3 inches from the root of the tail. Pieces for test shall be cut from only one side of the backbone of each skin.

4.6.1.2.2. Hides. The test area shall correspond to that shown in Figure 1. Pieces for test shall be cut from only one side of the backbone of each hide.

4.6.1.2.3. Sides, crops, backs, and bends. The test area shall correspond to that shown in Figure 1. Double bends and belting butts shall be sampled on only one side of the backbone.

4.6.1.2.4. Bellies. The test area shall correspond to area X of Figure 1. The middle of the area shall be approximately midway between D and E. If physical tests are required, the area may be enlarged to 5- by 7-inches.

4.6.1.2.5. Double shoulders. The test area shall correspond to area Y of Figure 1. The middle of the area shall be approximately midway between B and E. If physical tests are required, the area may be enlarged to 5- by 7-inches.

4.6.1.2.6. Pieces of bellies and shoulders for testing. When bellies and shoulders are purchased in separate units, pieces for use in chemical tests shall be taken from the areas marked "X" and "Y" respectively in Figure 1 and composited as specified in method ASTM D2813. When physical tests are required, the "X" and "Y" areas shall be enlarged to 5- by 7-inches and specimens for testing taken from that area.

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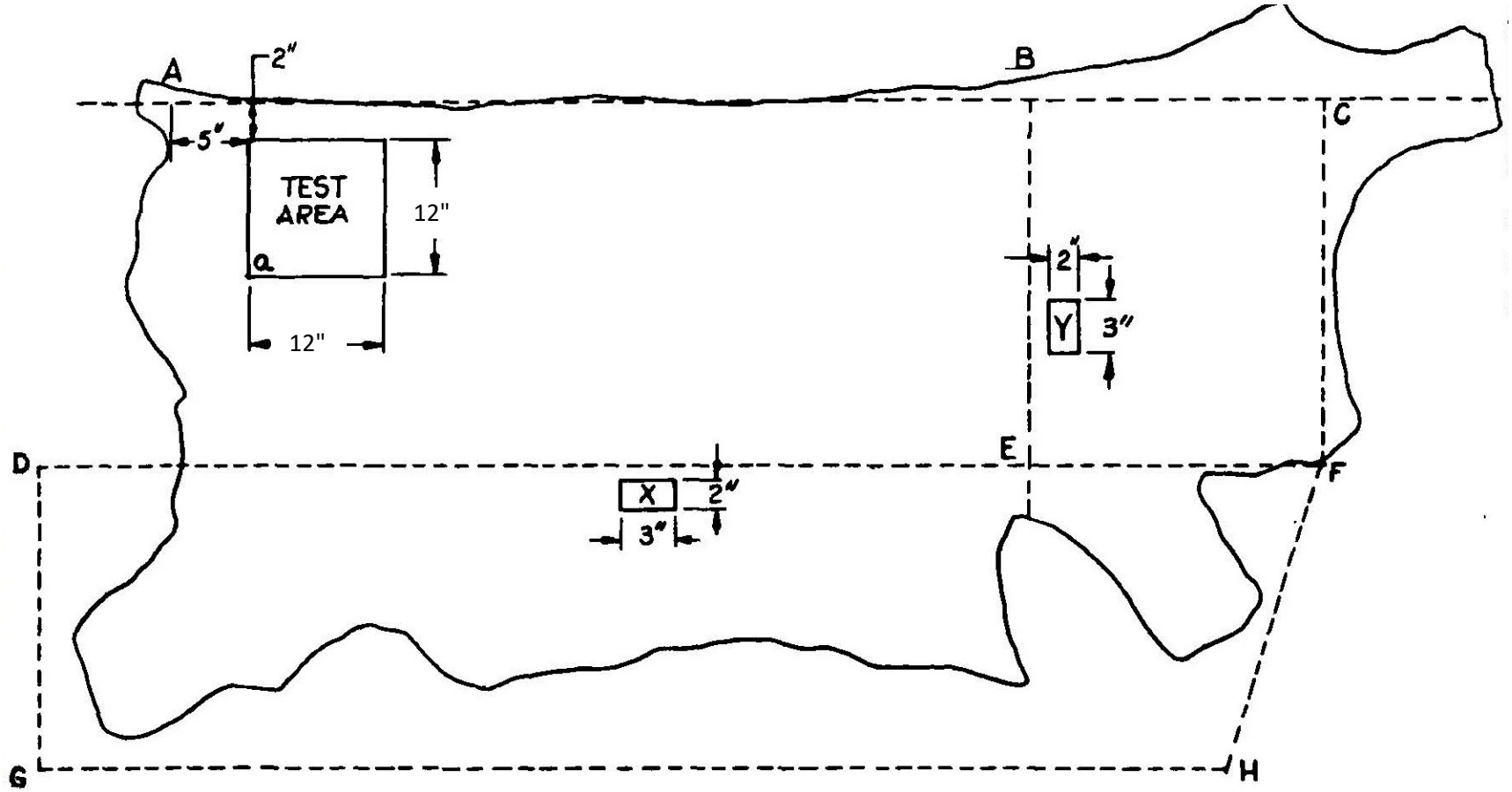


FIGURE 1. Location of test area.

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4.7. Sampling for testing of fabricated leather articles.

4.7.1. Sample size. Unless otherwise specified in the detail specification, the number of sample units to be taken from a lot for the purpose of physical and chemical tests shall be in accordance with Table II.

TABLE II. Number of sample units for physical and chemical tests.

Lot size (units)	Sample size
50 or less	2
51 to 500	3
501 to 3200	5
3201 and over	8

4.7.1.1. Sampling for testing physical characteristics. Unless otherwise specified in the material specification, specimens for each physical property for which requirements are specified shall be taken as described in the applicable test method.

4.7.1.2. Sampling for testing chemical properties. Unless otherwise specified in the material specification or applicable test method, chemical properties for which requirements are specified shall be determined on the composite sample that has been prepared in accordance with test method ASTM D2813.

4.8. Acceptance of lot.

4.8.1. Unless otherwise specified in the applicable procurement document, a lot shall be considered as meeting the requirements of that document if the following conditions are fulfilled:

(1) After examination of the lot, the number of defective units is less than the applicable rejection number.

(2) The results of physical and chemical test determinations performed on the sample unit or composite sample satisfy the requirements set forth in the procurement document.

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5. ATMOSPHERIC CONDITIONS FOR TESTING

5.1. Humidity and temperature conditions for testing. Unless otherwise specified in the applicable test method, material specification, or procurement document, strength and mechanical tests of leather and leather products shall be performed under Standard Atmospheric Conditions and performed on specimens in moisture equilibrium under Standard Atmospheric Conditions. All other types of tests may be performed under ambient conditions or as specified.

5.1.1. Standard Atmospheric Conditions and moisture equilibrium. Standard Atmospheric Conditions and moisture equilibrium for leather and leather products testing, shall be in accordance with ASTM D1610 at a 50 ± 4 percent relative humidity and at a temperature of 73.4 ± 3.6 °F (23 ± 2 °C).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1. Fungicide (mold resistance). If specified, the leather shall be treated with one of the following fungicides in Table III to provide mold resistance. The fungicides to treat the leather shall be an EPA registered product that complies with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA).

TABLE III. Fungicides.

Active Ingredient	Requirement (ppm)
2-(Thiocyanomethylthio) benzothiazole (TCMTB)	100 – 400
Diiodomethyl-*para*-tolylsulfone (DIMPTS)	300 – 2000

6.1.1. Verification of fungicide content is accomplished using test method ISO 13365-1 Leather – Chemical determination of the preservative (TCMTB, PCMC, OPP, OIT) content in leather by liquid chromatography – Part 1: Acetonitrile extraction method. When testing for DIMPTS, use an HPLC analytical standard to measure the concentration of DIMPTS in accordance with ISO 13365-1.

6.2. Numerical requirements in methods.

6.2.1. Forms used. Numerical requirements are given in any of three forms illustrated by the following examples: “approximately 2 grams,” “2 grams,” and “2.000 ± 0.002 grams.”

6.2.1.1. “Approximately 2 grams”. This form of expression implies that the numerical requirement is not critical and may vary within reason. The permissible variation is usually dictated by obvious practical considerations and the nearest readily obtained approximation to the weight or dimensions may be considered satisfactory.

6.2.1.2. “2 grams”. This form of expression implies that the numerical requirement is to be as close to “2 grams” as can be readily measured on the stated material with the usual, ordinary engineering tools.

6.2.1.3. “2.000 ± 0.002 grams”. This form of expression implies that the numerical requirement in question must be between 1.998 and 2.002 grams.

6.3. Raw materials of the tanning industry.

6.3.1. In describing various classes of leather, the name of the animal from which the skin was taken is generally used. The following are groups of animals from which skins are generally taken and dressed into leather:

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Cattle
Sheep and lamb
Goat and kid
Equine
Buffalo
Pig and hog
Deer
Kangaroo and wallaby
Aquatic
Miscellaneous

6.4. Finished leather, uses.

6.4.1. Cattle group.

6.4.1.1. Steer, cow, and bull hides produce leather for the following uses:

Shoe soles, heels, counters, welting, innersoles
Boot and shoe uppers
Harness, saddles, skirting for saddles, horse collars
Traveling bags, suitcases, straps
Gloves and garments
Machinery belting
Upholstery for automobiles, furniture, airplanes, and decorative purposes
Fancy leather goods, handbags, belts, and wallets
Bookbinding's
Aprons
Buffing wheels
Aprons for carders, combers, and pickers on textile machinery
Hydraulic packings, washers, and gaskets
Laces for belting, shoes, parachute slings, and tags
Lithographic purposes
Razor strops
Suspenders
Rawhides
Self-sealing gasoline tanks
Briefcases, envelopes (slide fastener)
Orthopedic purposes
Reinforcements for frame baskets

6.4.1.2. Kipskins from large calves or undersized or small breeds of cattle produce leather for the following uses:

Shoe uppers
Fancy leather goods and luggage
Gloves and garments
Shoe linings

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6.4.1.3. Calfskins produce leather for the following uses:

Boot and shoe uppers
Fancy leather goods and handbags
Gloves and garments
Sweatbands for hats
Rawhide and parchment
Military helmets and gas masks
Fine shoe linings
Bookbindings
Grips for golf clubs
Handicrafts

6.4.2. Sheep and lamb group.

6.4.2.1. Woolen skins, hair skins (cabrettas), etc., produce leather for the following uses:

Shoe linings and uppers, slippers, and play shoes
Gloves
Bookbinding's
Leather goods and handbags
Aprons
Chamois
Coats and other articles of clothing
Aviators' clothing
Sweatbands for hats
Meters
Parchment
Piano actions
Rollers on textile machinery
Military helmets
Millinery and caps
Sporting goods

6.4.3. Goat and kid group.

6.4.3.1. Skins from this group produce leather for the following uses:

Shoe uppers
Fancy leather goods and handbags
Gloves and garments
Aviators' clothing
Bookbinding's
Upholstery

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6.4.4. Equine group.

6.4.4.1. Horse, colt, ass, mule, and zebra hides produce leather for the following uses:

Shoes
Gloves and garments
Aviators' clothing
Sporting goods (baseball covers and mitts)
Puttees
Luggage
Razor strops
Belts

6.4.5. Buffalo group.

6.4.5.1. Domesticated land and water buffalo (excluding American bison) produce leather for the following uses:

Shoe soles and uppers
Buffing wheels
Luggage
Handbags

6.4.6. Pig and hog group.

6.4.6.1. Pig, hog, boar, peccary, and carpincho (Brazilian rodent) skins produce leather for the following uses:

Fancy leather goods and luggage
Gloves
Innersoles, counters, etc.
Saddlery and harness
Shoe uppers

6.4.7. Deer group.

6.4.7.1. Fallow deer, reindeer, antelope, gazelle (dikdik), elk, and caribou skins produce leather for the following uses:

Shoe uppers
Gloves
Clothing
Fancy leather goods
Piano actions
Mukluks

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6.4.8. Kangaroo and wallaby group.

6.4.8.1. Skins from this group produce leather for shoe uppers.

6.4.9. Aquatic group.

6.4.9.1. Seal, sea lion, and walrus skins produce leather for the following uses:

Luggage
Fancy leather goods
Buffing wheels

6.4.9.2. Shark, whale, blackfish, dolphin, porpoise, etc., produce leather for the following uses:

Fancy leather goods
Luggage
Shoe uppers

6.4.9.3. Alligator skins produce leather for the following uses:

Shoes
Handbags
Luggage
Fancy leather goods

6.4.10. Miscellaneous group.

6.4.10.1. Ostrich skins produce leather for the following uses:

Fancy leather goods
Luggage

6.4.10.2. Camel, llama, alpaca, and yak skins are tanned in the producing countries usually for local consumption.

6.4.10.3. Lizard, snake, frog, and similar skins produce leather for the following uses:

Shoe uppers
Fancy leather goods

6.4.10.4. Elephant, hippopotamus, and rhinoceros' skins are not used to any great extent because of small supply and the long time required to tan it.

6.5. Intended use. To provide standardized test procedures for evaluating quality of leather that are used in military footwear, handwear, and other leather products.

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6.6. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this standard.
b. Test method required.

6.7. Source of Non-Government documents.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Evaluation Procedures:
AATCC EP8 - AATCC 9-Step Chromatic Transference Scale

(Copies of this document are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

ASTM INTERNATIONAL (ASTM)

ASTM D1516 - Width of Leather
ASTM D1517 - Terminology Relating to Leather
ASTM D1611 - Corrosion Produced by Leather in Contact with Metal
ASTM D1813 - Measuring Thickness of Leather Test Specimens
ASTM D1814 - Measuring Thickness of Leather Units
ASTM D2098 - Dynamic Water Resistance of Shoe Upper Leather by the Dow Corning Leather Tester
ASTM D2099 - Dynamic Water Resistance of Shoe Upper Leather by the Maeser Water Penetration Tester
ASTM D2207 - Bursting Strength of Leather by the Ball Method
ASTM D2208 - Breaking Strength of Leather by the Grab Method
ASTM D2209 - Tensile Strength of Leather
ASTM D2210 - Grain Crack and Extension of Leather by the Mullen Test
ASTM D2212 - Slit Tear Resistance of Leather
ASTM D2213 - Compressibility of Leather
ASTM D2322 - Resistance of Shoe Upper Leather to Artificial Perspiration
ASTM D2346 - Apparent Density of Leather
ASTM D2617 - Total Ash in Leather
ASTM D2807 - Chromic Oxide in Leather (Perchloric Acid Oxidation)
ASTM D2810 - pH of Leather
ASTM D2813 - Sampling Leather for Physical and Chemical Tests
ASTM D2821 - Measuring the Relative Stiffness of Leather by Means of a Torsional Wire Apparatus
ASTM D2868 - Nitrogen Content (Kjeldahl) and Hide Substance Content of Leather, Wet Blue and Wet White

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ASTM D2941 - Measuring Break Pattern of Leather (Break Scale)
ASTM D3495 - Hexane Extraction of Leather
ASTM D3790 - Volatile Matter (Moisture) of Leather by Oven Drying
ASTM D4704 - Tearing Strength, Tongue Tear of Leather
ASTM D4705 - Stitch Tear Strength of Leather, Double Hole
ASTM D4786 - Stitch Tear Strength, Single Hole
ASTM D4831 - Buckle Tear Strength of Leather
ASTM D5052 - Permeability of Leather to Water Vapor
ASTM D5053 - Colorfastness of Crocking of Leather
ASTM D6012 - Determination of Resistance of Leather to (Bleeding) Color Stain Transfer
ASTM D6013 - Determination of Area Stability of Leather to Laundering
ASTM D6014 - Determination of Dynamic Water Absorption of Leather Surfaces
ASTM D6015 - Static Water Absorption of Leather
ASTM D6016 - Determination of Nitrogen, Water Extractable in Leather
ASTM D6017 - Determination of Magnesium Sulfate (Epsom Salt) in Leather
ASTM D6020 - Calculation of (Non-Mineral) Combined Tanning Agents and Degree of Tannage
ASTM D6075 - Cracking Resistance of Leather
ASTM D6076 - Shrinkage Temperature of Leather
ASTM D6116 - Blocking
ASTM D6183 - Tackiness of Finish on Leather
ASTM D7912 - Resistance of Finish to Heat Aging (Finish Stability)

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 13365-1 - Leather – Chemical determination of the preservative (TCMTB, PCMC, OPP, OIT) content in leather by liquid chromatography – Part 1: Acetonitrile extraction method
ISO 17072-2 - Leather – Chemical determination of metal content – Part 2: Total metal content

(Copies of these documents are available online at https://www.iso.org.)

6.8. Subject term (key word) listing.

Footwear
Gloves
Inspection
Procedures

6.9. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTVITY:
GSA-FAS

Custodians:
Army – GL
Navy – SA
Air Force – 11
DLA – CT

Preparing activity:
DLA – CT
(Project 83GP-2024-004)

Review activities:
Navy – AS, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil.

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